

Unlabeled and Labeled Graphs

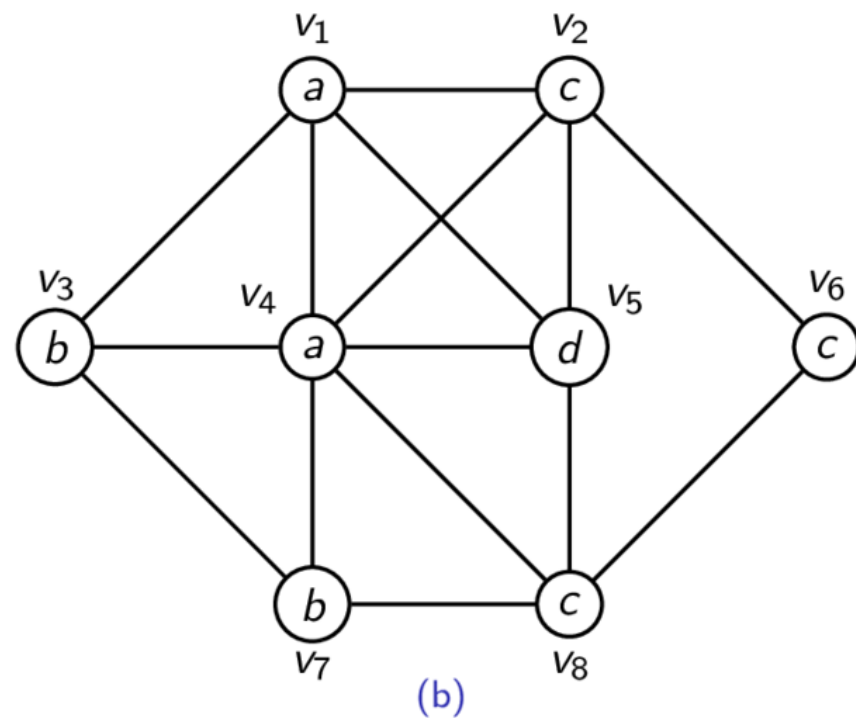
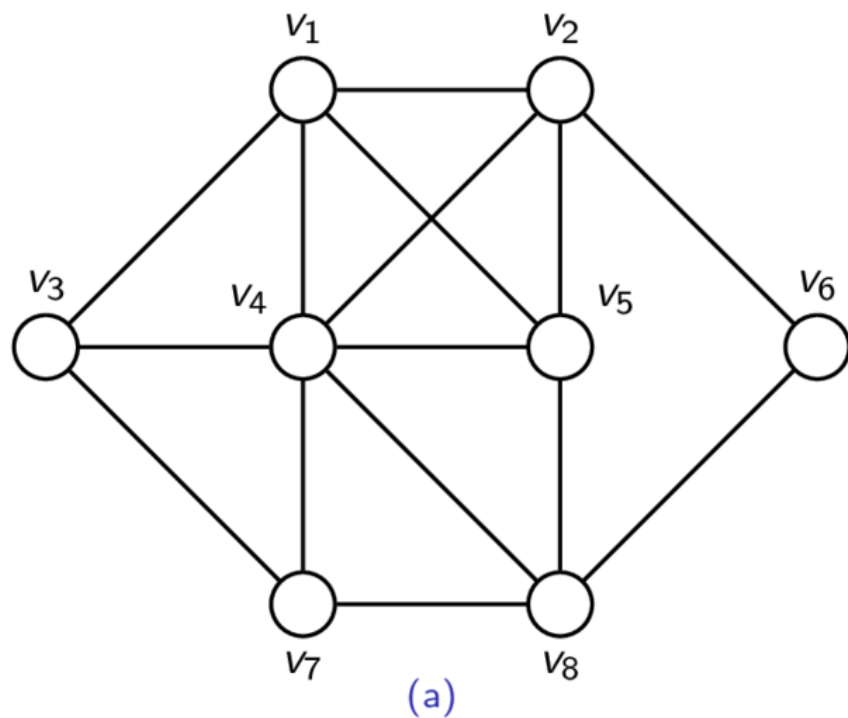
The goal of graph mining is to extract interesting subgraphs from a single large graph (e.g., a social network), or from a database of many graphs.

A graph is a pair $G = (V, E)$ where V is a set of vertices, and $E \subseteq V \times V$ is a set of edges.

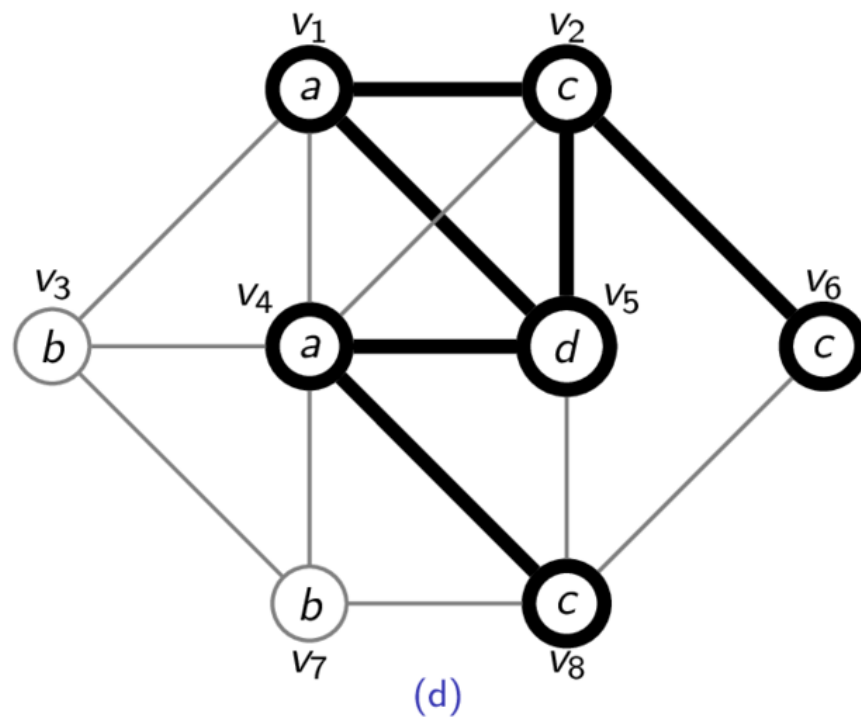
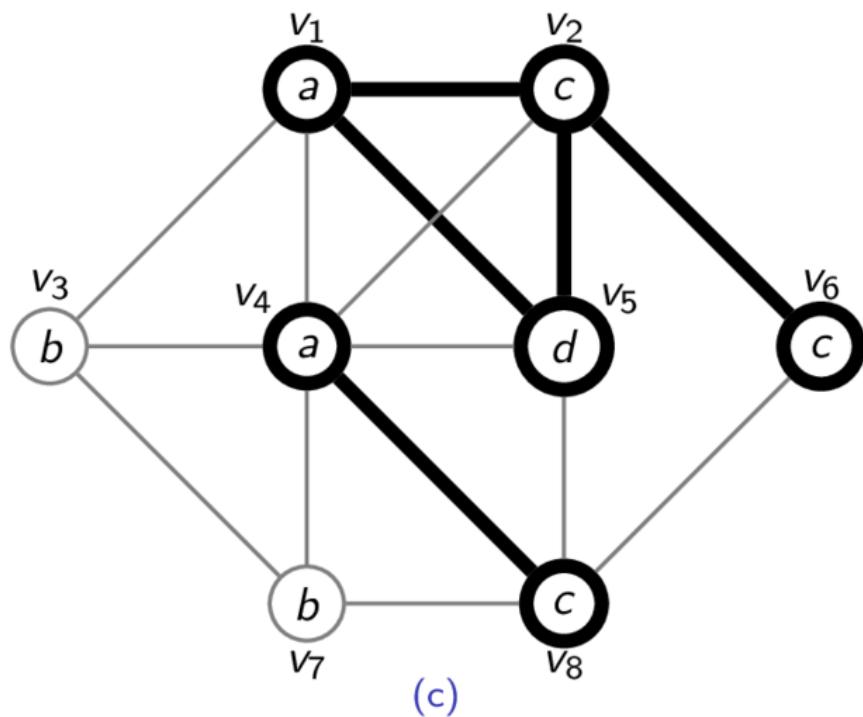
A *labeled graph* has labels associated with its vertices as well as edges. We use $L(u)$ to denote the label of the vertex u , and $L(u, v)$ to denote the label of the edge (u, v) , with the set of vertex labels denoted as Σ_V and the set of edge labels as Σ_E . Given an edge $(u, v) \in G$, the tuple $\langle u, v, L(u), L(v), L(u, v) \rangle$ that augments the edge with the node and edge labels is called an *extended edge*.

A graph $G' = (V', E')$ is said to be a *subgraph* of G if $V' \subseteq V$ and $E' \subseteq E$. A *connected subgraph* is defined as a subgraph G' such that $V' \subseteq V$, $E' \subseteq E$, and for any two nodes $u, v \in V'$, there exists a *path* from u to v in G' .

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Subgraph and Connected Subgraph



Mining Frequent Subgraph

Given a database of graphs, $\mathbf{D} = \{G_1, G_2, \dots, G_n\}$, and given some graph G , the support of G in $\mathbf{D}$ is defined as follows:

$$\text{sup}(G) = \left| \{G_i \in \mathbf{D} \mid G \subseteq G_i\} \right|$$

The support is simply the number of graphs in the database that contain G . Given a *minsup* threshold, the goal of graph mining is to mine all frequent connected subgraphs with $\text{sup}(G) \geq \text{minsup}$.

If we consider subgraphs with m vertices, then there are $\binom{m}{2} = O(m^2)$ possible edges. The number of possible subgraphs with m nodes is then $O(2^{m^2})$ because we may decide either to include or exclude each of the edges. Many of these subgraphs will not be connected, but $O(2^{m^2})$ is a convenient upper bound. When we add labels to the vertices and edges, the number of labeled graphs will be even more.

Graph Pattern Mining

There are two main challenges in frequent subgraph mining.

The first is to systematically generate nonredundant candidate subgraphs. We use *edge-growth* as the basic mechanism for extending the candidates.

The second challenge is to count the support of a graph in the database. This involves subgraph isomorphism checking, as we have to find the set of graphs that contain a given candidate.